

STANDARD OPERATING PROCEDURE

SOP-MECH-021 Rev.2 - Alignment of Centrifugal Pumps and Drivers

Document No.: SOP-MECH-021

Revision: 2

Effective Date: 15 Jun 2023

Custodian: Head - Mechanical Maintenance

1. Scope

This procedure covers cold alignment of horizontal centrifugal pumps to their drivers using reverse dial indicator or laser alignment methods. It applies to all pumps within the refinery of frame size 2 and above.

2. Alignment Tolerances

| Parameter            | Acceptable         | Alert              | Unacceptable          | Method         |
|----------------------|--------------------|--------------------|-----------------------|----------------|
| Angular misalignment | 0.05 mm per 100 mm | 0.08 mm per 100 mm | Above 0.10 mm per 100 | Dial or laser  |
| Parallel offset      | 0.05 mm            | 0.10 mm            | Above 0.15 mm         | Dial or laser  |
| Soft foot            | 0.05 mm            | 0.08 mm            | Above 0.10 mm         | Dial indicator |
| Shaft run out        | 0.03 mm            | 0.05 mm            | Above 0.08 mm         | Dial indicator |

Soft foot shall be corrected to within 0.05 mm before any alignment readings are taken. Attempting to align a machine with an uncorrected soft foot produces readings that cannot be repeated.

3. Post Alignment Vibration Acceptance

| Machine Class                  | Good         | Alarm        | Trip          | Standard    |
|--------------------------------|--------------|--------------|---------------|-------------|
| Centrifugal pump, below 15 kW  | 2.8 mm/s RMS | 4.5 mm/s RMS | 7.1 mm/s RMS  | ISO 10816-3 |
| Centrifugal pump, 15 to 300 kW | 4.5 mm/s RMS | 7.1 mm/s RMS | 11.0 mm/s RMS | ISO 10816-3 |
| Motor, rigid mounted           | 2.8 mm/s RMS | 4.5 mm/s RMS | 7.1 mm/s RMS  | ISO 10816-3 |

The vibration alarm limit for centrifugal pumps in the 15 to 300 kW range is 7.1 mm/s RMS measured at the bearing housing in the horizontal direction. A machine exceeding the alarm limit shall be reported to the Reliability Engineer within one shift.

- 4.1 All work shall be carried out under a valid work permit issued by the area authority. The permit shall remain displayed at the work site for the duration of the activity and shall be returned on completion.
- 4.2 Personnel engaged in this activity shall have completed the department competency assessment within the preceding twenty four months. Records of assessment shall be available for audit.
- 4.3 Any deviation from this procedure shall be recorded on the deviation register and shall be approved in writing by the Head of Inspection before work proceeds. Verbal approval is not acceptable.
- 4.4 Calibration certificates for all test instruments shall be current and traceable to a national standard. An instrument whose calibration has lapsed shall be withdrawn from service immediately.

Prepared By

Reviewed By

Approved By